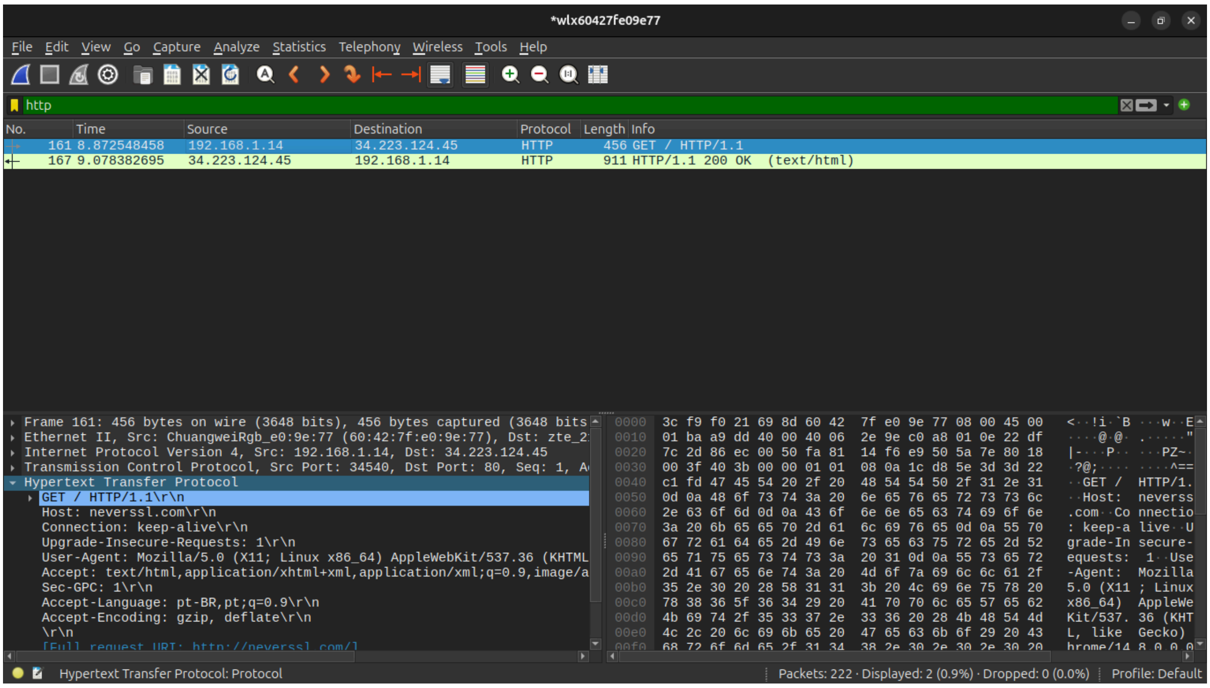
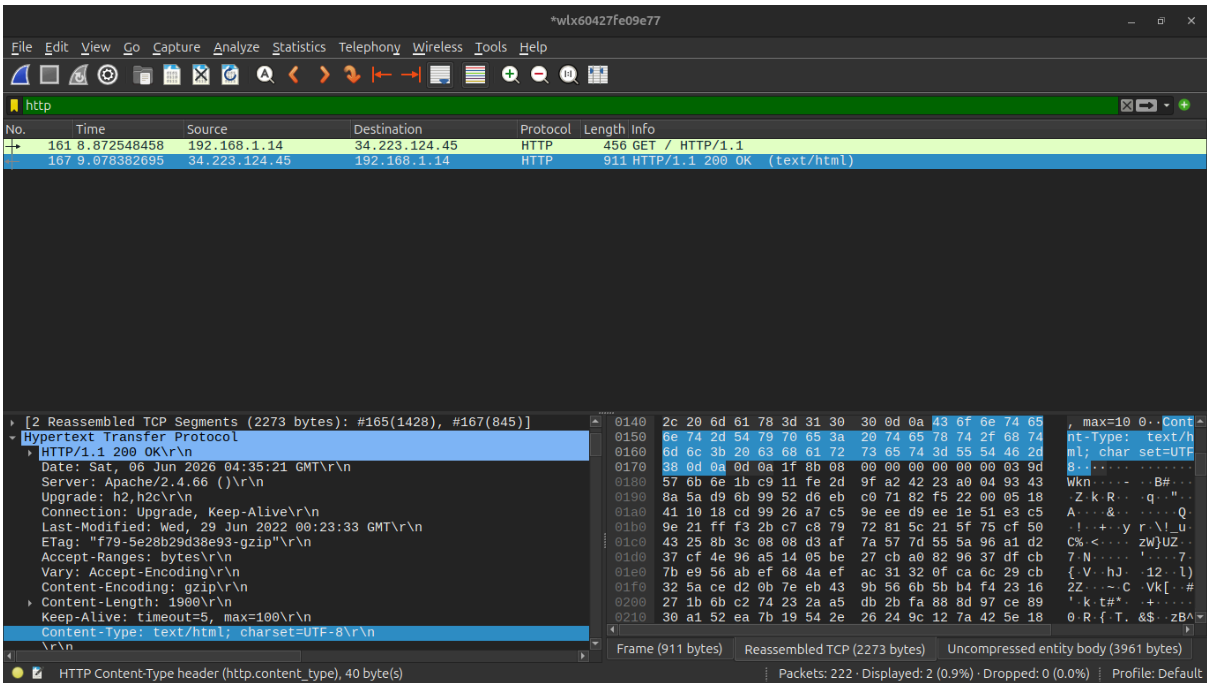
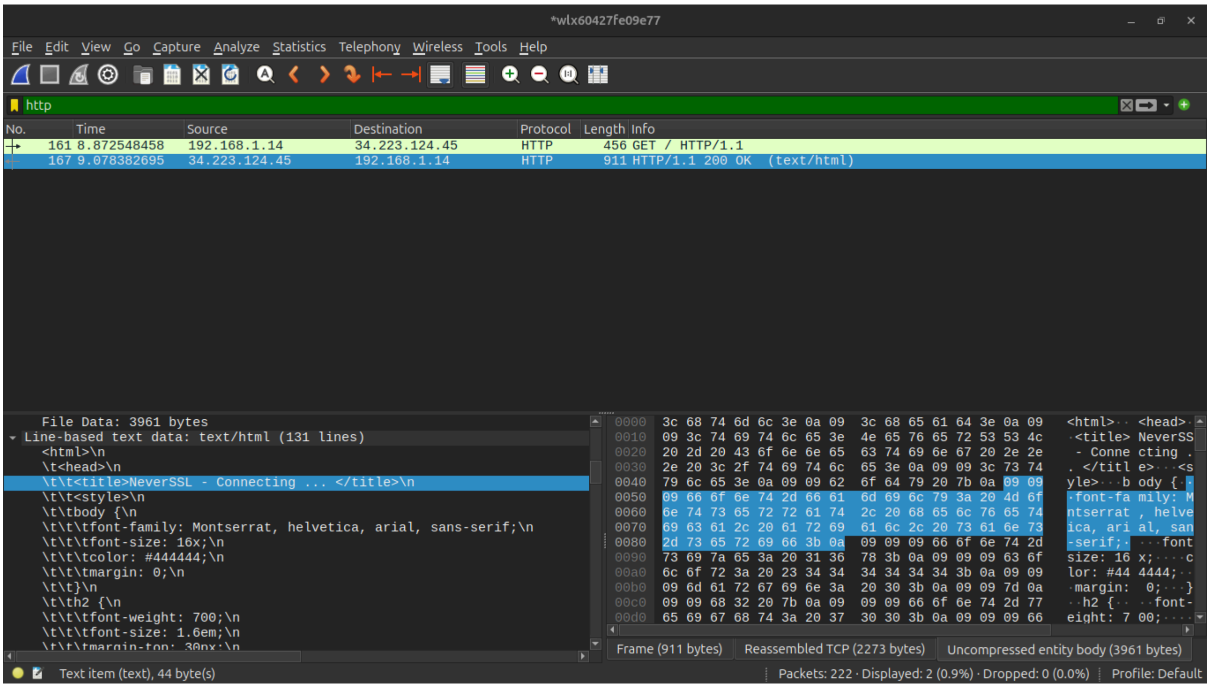
Exercício 5

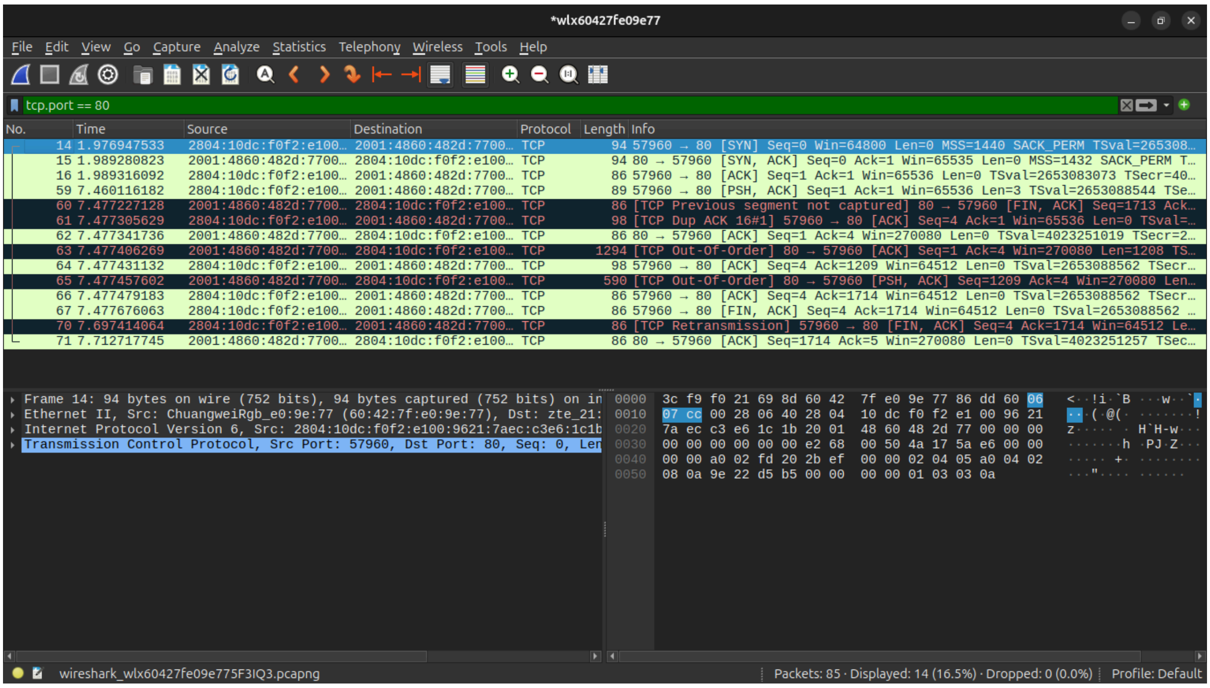


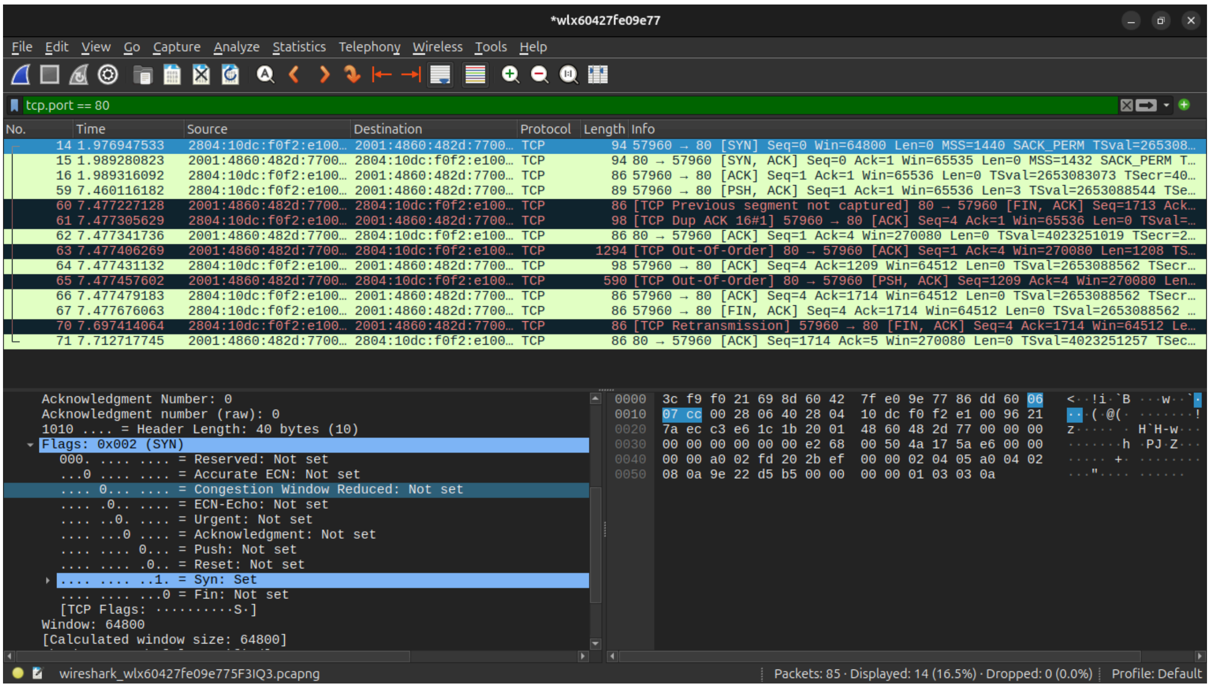


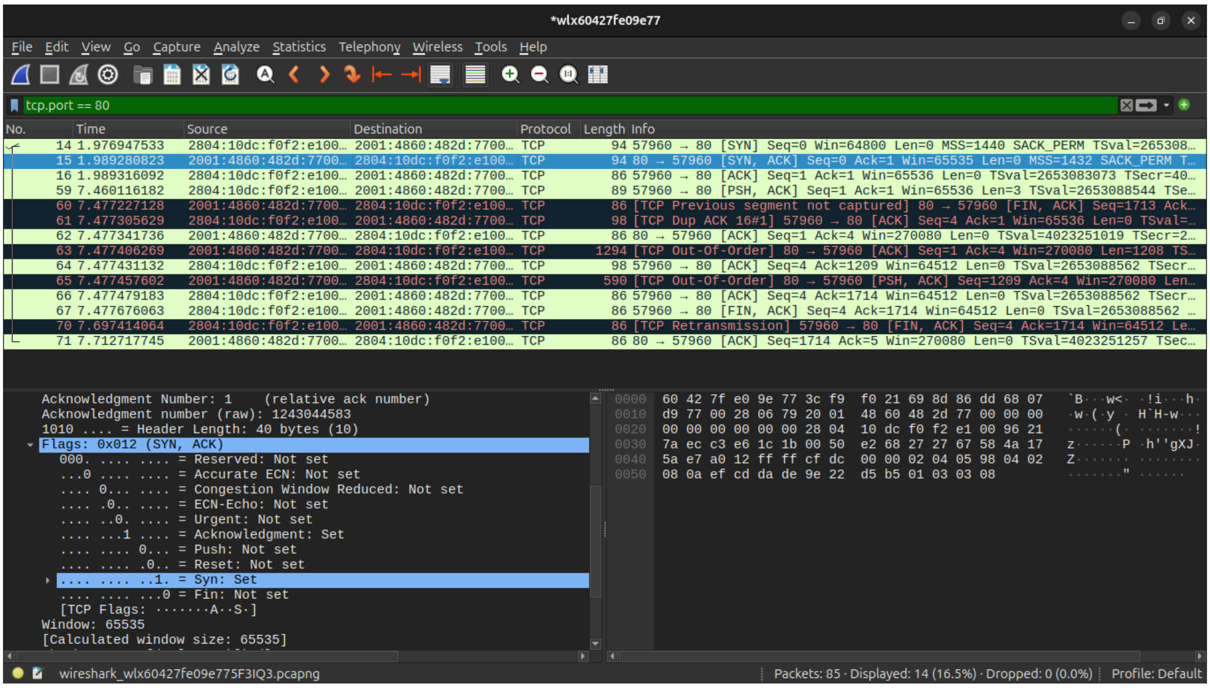


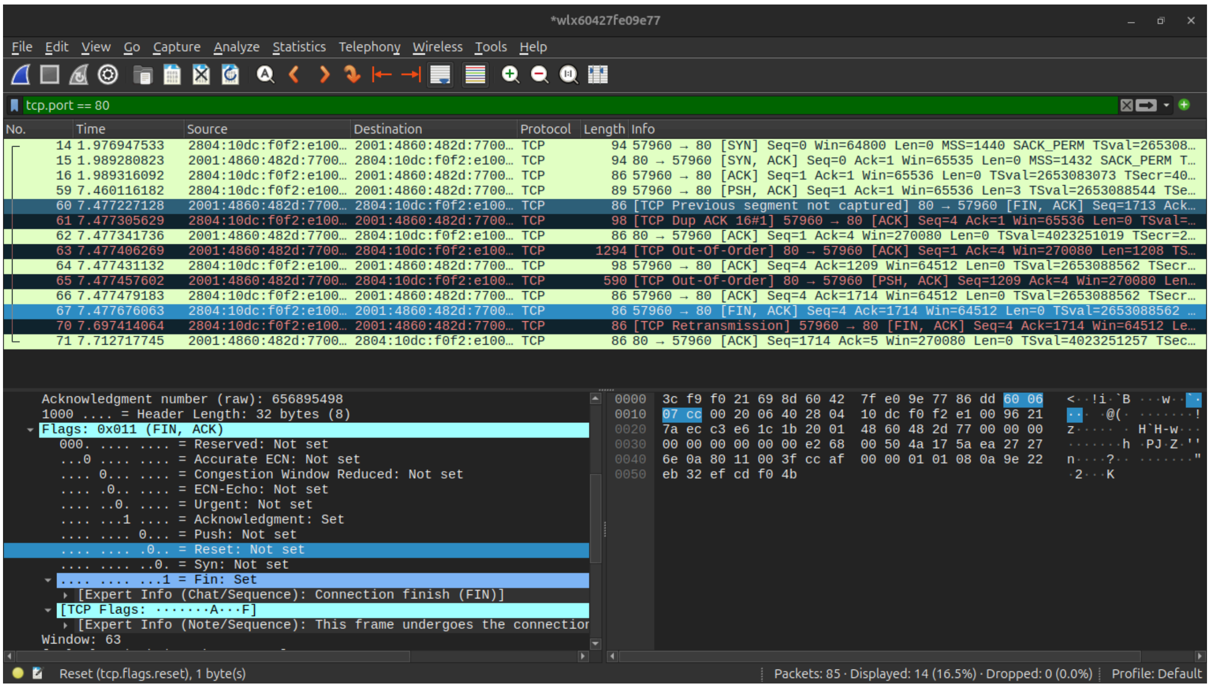
A requisição utilizou o metodo GET para o Host neverssl.com pelo navegador Mozilla. A resposta do servidor foi um HTTP 200 OK na data 6 de jun de 2026, o server usa Apache, a ultima modificação foi em 29 de junho de 2022 o content-Type é text/html indicando que é uma pagina web. Logo abaixo das linhas de cabeçalho, tem uma seção chamada "Line-based text data: text/html" onde é possível ver o html dessa página.

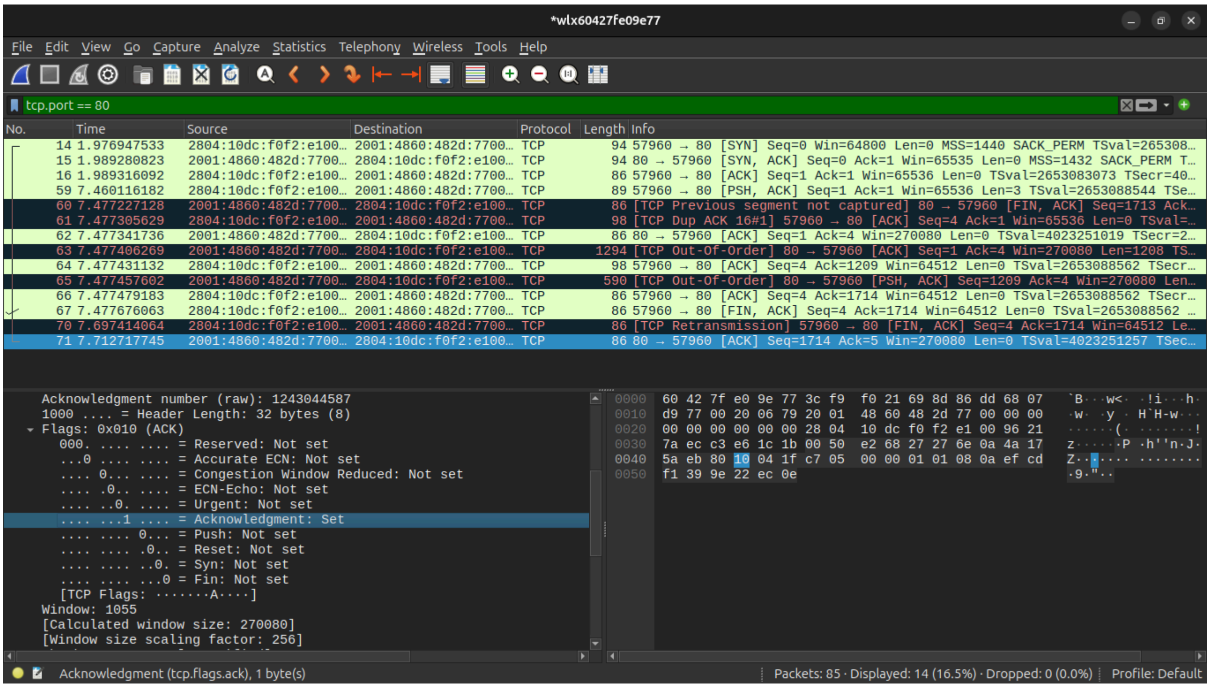
Exercício 6




*wlx60427fe09e77
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.port == 80
No. Time Source Destination Protocol Length Info
14 1.976947533 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 94 57960 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=265308…
15 1.989280823 2001:4860:482d:7700… 2804:10dc:f0f2:e100… TCP 94 80 → 57960 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1432 SACK_PERM T…
16 1.989316092 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 86 57960 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=2653083073 TSecr=40…
59 7.460116182 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 89 57960 → 80 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=3 TSval=2653088544 TSe…
60 7.477227128 2001:4860:482d:7700… 2804:10dc:f0f2:e100… TCP 86 [TCP Previous segment not captured] 80 → 57960 [FIN, ACK] Seq=1713 Ack…
61 7.477305629 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 98 [TCP Dup ACK 16#1] 57960 → 80 [ACK] Seq=4 Ack=1 Win=65536 Len=0 TSval=…
62 7.477341736 2001:4860:482d:7700… 2804:10dc:f0f2:e100… TCP 86 80 → 57960 [ACK] Seq=1 Ack=4 Win=270080 Len=0 TSval=4023251019 TSecr=2…
63 7.477406269 2001:4860:482d:7700… 2804:10dc:f0f2:e100… TCP 1294 [TCP Out-Of-Order] 80 → 57960 [ACK] Seq=1 Ack=4 Win=270080 Len=1208 TS…
64 7.477431132 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 98 57960 → 80 [ACK] Seq=4 Ack=1209 Win=64512 Len=0 TSval=2653088562 TSecr…
65 7.477457602 2001:4860:482d:7700… 2804:10dc:f0f2:e100… TCP 590 [TCP Out-Of-Order] 80 → 57960 [PSH, ACK] Seq=1209 Ack=4 Win=270080 Len…
66 7.477479183 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 86 57960 → 80 [ACK] Seq=4 Ack=1714 Win=64512 Len=0 TSval=2653088562 TSecr…
67 7.477676063 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 86 57960 → 80 [FIN, ACK] Seq=4 Ack=1714 Win=64512 Len=0 TSval=2653088562 …
70 7.697414064 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 86 [TCP Retransmission] 57960 → 80 [FIN, ACK] Seq=4 Ack=1714 Win=64512 Le…
71 7.712717745 2001:4860:482d:7700… 2804:10dc:f0f2:e100… TCP 86 80 → 57960 [ACK] Seq=1714 Ack=5 Win=270080 Len=0 TSval=4023251257 TSec…
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1010 .... = Header Length: 40 bytes (10)
Flags: 0x002 (SYN)
000. .... = Reserved: Not set
...0 .... = Accurate ECN: Not set
.... 0... .... = Congestion Window Reduced: Not set
.... .0.. .... = ECN-Echo: Not set
.... ..0. .... = Urgent: Not set
....0 .... = Acknowledgment: Not set
.... 0... = Push: Not set
.... .0.. = Reset: Not set
....1. = Syn: Set
....0 = Fin: Not set
[TCP Flags: ··········S·]
Window: 64800
[Calculated window size: 64800]
0000 3c f9 f0 21 69 8d 60 42 7f e0 9e 77 86 dd 60 06
0010 07 cc 00 28 06 40 28 04 10 dc f0 f2 e1 00 96 21
0020 7a ec c3 e6 1c 1b 20 01 48 60 48 2d 77 00 00 00
0030 00 00 00 00 00 00 e2 68 00 50 4a 17 5a e6 00 00
0040 00 00 a0 02 fd 20 2b ef 00 00 02 04 05 a0 04 02
0050 08 0a 9e 22 d5 b5 00 00 00 00 01 03 03 0a
wireshark_wlx60427fe09e775F3IQ3.pcapng
Packets: 85 · Displayed: 14 (16.5%) · Dropped: 0 (0.0%)
Profile: Default

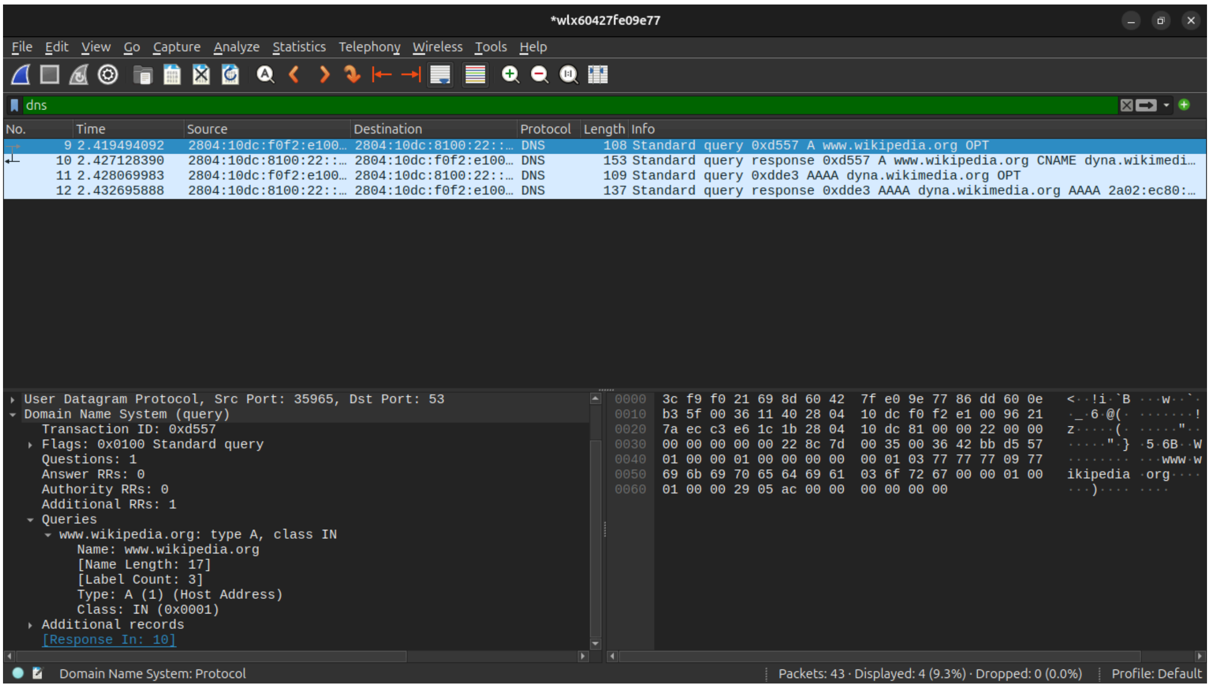

*wlx60427fe09e77
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.port == 80
No. Time Source Destination Protocol Length Info
14 1.976947533 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 94 57960 → 80 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=265308…
15 1.989280823 2001:4860:482d:7700… 2804:10dc:f0f2:e100… TCP 94 80 → 57960 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1432 SACK_PERM T…
16 1.989316092 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 86 57960 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=2653083073 TSecr=40…
59 7.460116182 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 89 57960 → 80 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=3 TSval=2653088544 TSe…
60 7.477227128 2001:4860:482d:7700… 2804:10dc:f0f2:e100… TCP 86 [TCP Previous segment not captured] 80 → 57960 [FIN, ACK] Seq=1713 Ack…
61 7.477305629 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 98 [TCP Dup ACK 16#1] 57960 → 80 [ACK] Seq=4 Ack=1 Win=65536 Len=0 TSval=…
62 7.477341736 2001:4860:482d:7700… 2804:10dc:f0f2:e100… TCP 86 80 → 57960 [ACK] Seq=1 Ack=4 Win=270080 Len=0 TSval=4023251019 TSecr=2…
63 7.477406269 2001:4860:482d:7700… 2804:10dc:f0f2:e100… TCP 1294 [TCP Out-Of-Order] 80 → 57960 [ACK] Seq=1 Ack=4 Win=270080 Len=1208 TS…
64 7.477431132 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 98 57960 → 80 [ACK] Seq=4 Ack=1209 Win=64512 Len=0 TSval=2653088562 TSecr…
65 7.477457602 2001:4860:482d:7700… 2804:10dc:f0f2:e100… TCP 590 [TCP Out-Of-Order] 80 → 57960 [PSH, ACK] Seq=1209 Ack=4 Win=270080 Len…
66 7.477479183 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 86 57960 → 80 [ACK] Seq=4 Ack=1714 Win=64512 Len=0 TSval=2653088562 TSecr…
67 7.477676063 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 86 57960 → 80 [FIN, ACK] Seq=4 Ack=1714 Win=64512 Len=0 TSval=2653088562 …
70 7.697414064 2804:10dc:f0f2:e100… 2001:4860:482d:7700… TCP 86 [TCP Retransmission] 57960 → 80 [FIN, ACK] Seq=4 Ack=1714 Win=64512 Le…
71 7.712717745 2001:4860:482d:7700… 2804:10dc:f0f2:e100… TCP 86 80 → 57960 [ACK] Seq=1714 Ack=5 Win=270080 Len=0 TSval=4023251257 TSec…
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1243044583
1010 .... = Header Length: 40 bytes (10)
Flags: 0x012 (SYN, ACK)
000. .... = Reserved: Not set
...0 .... = Accurate ECN: Not set
.... 0... .... = Congestion Window Reduced: Not set
.... .0.. .... = ECN-Echo: Not set
.... ..0. .... = Urgent: Not set
....1 .... = Acknowledgment: Set
.... 0... = Push: Not set
.... .0.. = Reset: Not set
....1. = Syn: Set
....0 = Fin: Not set
[TCP Flags: ·······A··S·]
Window: 65535
[Calculated window size: 65535]
0000 60 42 7f e0 9e 77 3c f9 f0 21 69 8d 86 dd 68 07
0010 d9 77 00 28 06 79 20 01 48 60 48 2d 77 00 00 00
0020 00 00 00 00 00 00 28 04 10 dc f0 f2 e1 00 96 21
0030 7a ec c3 e6 1c 1b 00 50 e2 68 27 27 67 58 4a 17
0040 5a e7 a0 12 ff ff cf dc 00 00 02 04 05 98 04 02
0050 08 0a ef cd da de 9e 22 d5 b5 01 03 03 08
wireshark_wlx60427fe09e775F3IQ3.pcapng
Packets: 85 · Displayed: 14 (16.5%) · Dropped: 0 (0.0%)
Profile: Default

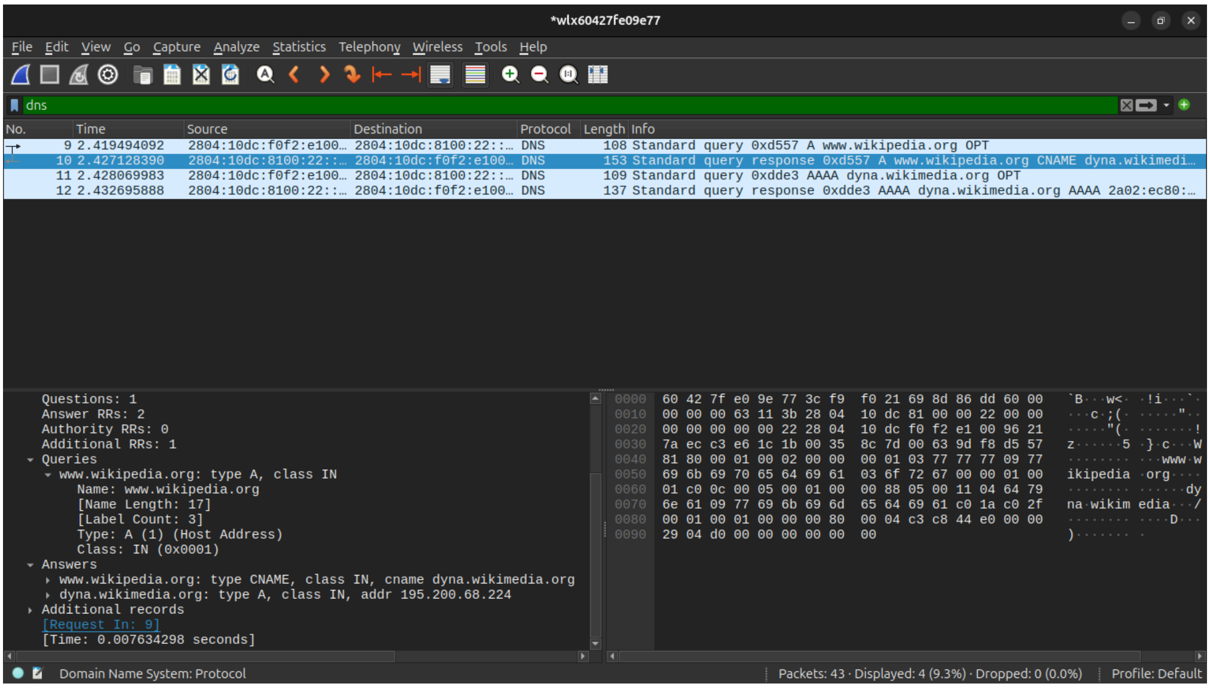


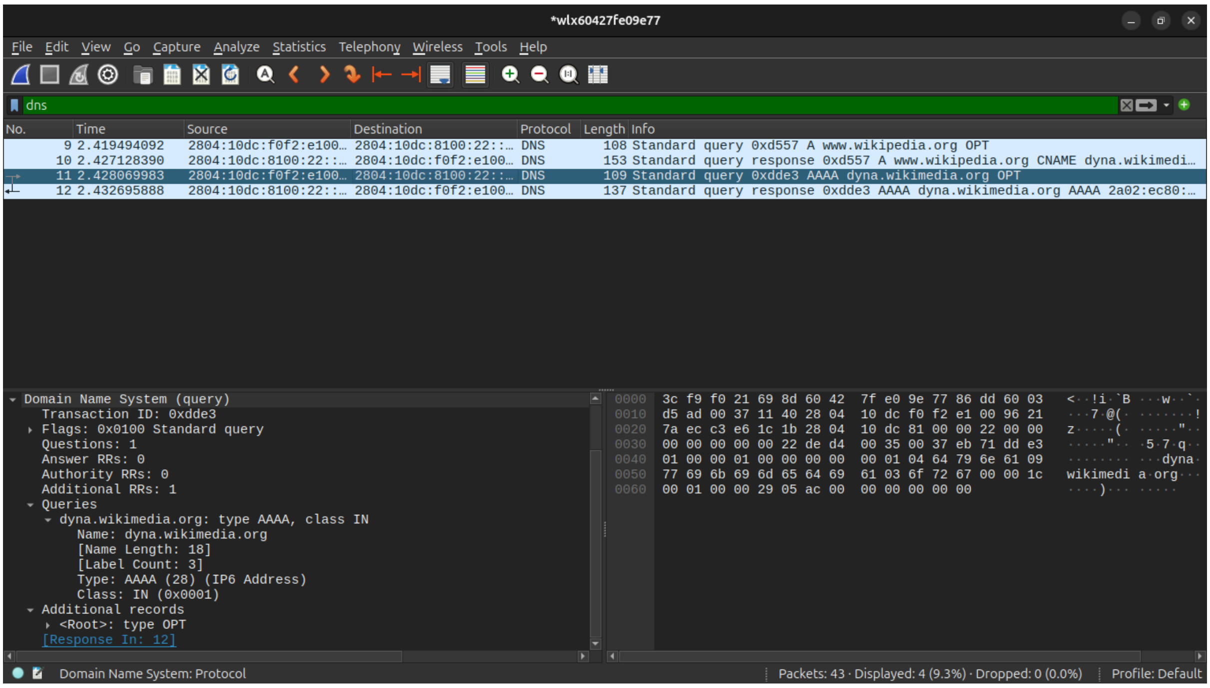


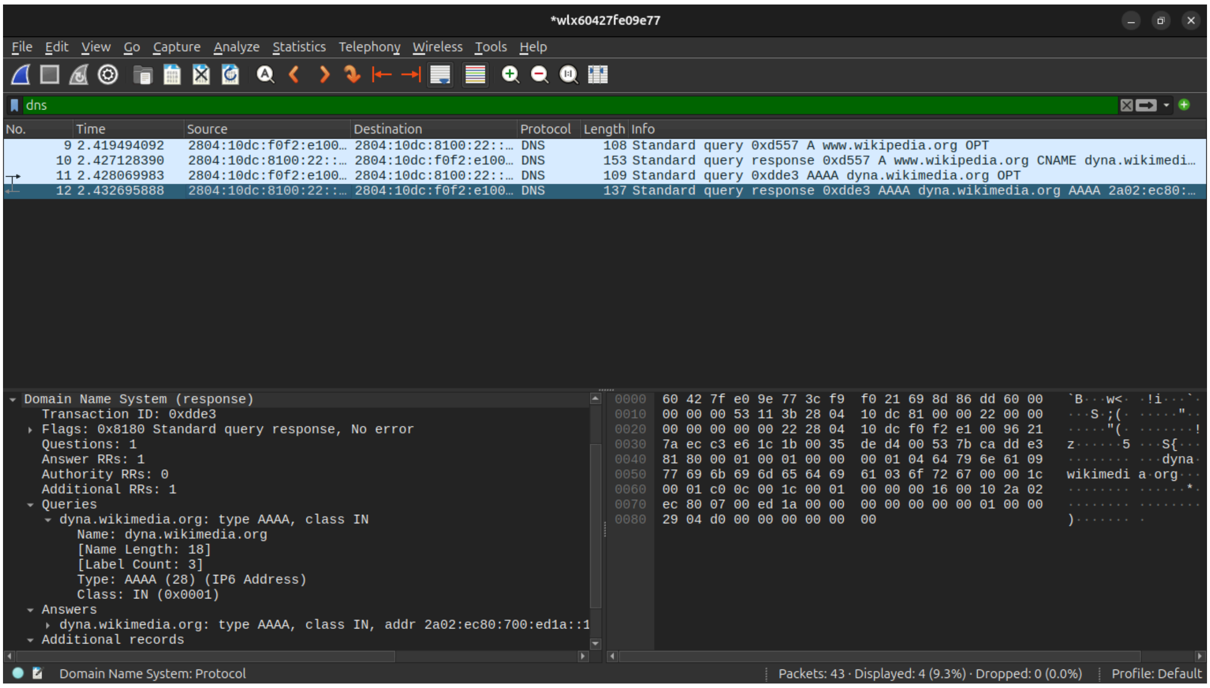
Na primeira linha é possível ver o envio do flag SYN para fazer a conexão com a porta 80 do servidor da Google, expandindo as flags podemos ver que a flag Syn está marcada com 1. Na segunda linha é possível ver que o servidor respondeu com as flags Syn e Acknowledgment marcados com 1 e logo em seguida é enviado uma flag ACK do meu computador para o servidor da Google, assim estabelecendo a conexão entre os dois. No encerramento da conexão onde digitei uma tecla qualquer e apertei enter, assim o meu computador enviou com as flags FIN e ACK marcados com 1 e esse pacote foi retransmitindo logo abaixo e então teve a resposta do servidor da Google com o flag ACK marcado com 1.

Exercício 7


*wlx60427fe09e77
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Length Info
9 2.419494092 2804:10dc:f0f2:e100… 2804:10dc:8100:22::… DNS 108 Standard query 0xd557 A www.wikipedia.org OPT
10 2.427128390 2804:10dc:8100:22::… 2804:10dc:f0f2:e100… DNS 153 Standard query response 0xd557 A www.wikipedia.org CNAME dyna.wikimedi…
11 2.428069983 2804:10dc:f0f2:e100… 2804:10dc:8100:22::… DNS 109 Standard query 0xdde3 AAAA dyna.wikimedia.org OPT
12 2.432695888 2804:10dc:8100:22::… 2804:10dc:f0f2:e100… DNS 137 Standard query response 0xdde3 AAAA dyna.wikimedia.org AAAA 2a02:ec80:…
User Datagram Protocol, Src Port: 35965, Dst Port: 53
Domain Name System (query)
Transaction ID: 0xd557
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
Queries
www.wikipedia.org: type A, class IN
Name: www.wikipedia.org
[Name Length: 17]
[Label Count: 3]
Type: A (1) (Host Address)
Class: IN (0x0001)
Additional records
[Response In: 10]
0000 3c f9 f0 21 69 8d 60 42 7f e0 9e 77 86 dd 60 0e
0010 b3 5f 00 36 11 40 28 04 10 dc f0 f2 e1 00 96 21
0020 7a ec c3 e6 1c 1b 28 04 10 dc 81 00 00 22 00 00
0030 00 00 00 00 00 22 8c 7d 00 35 00 36 42 bb d5 57
0040 01 00 00 01 00 00 00 00 00 01 03 77 77 77 09 77
0050 69 6b 69 70 65 64 69 61 03 6f 72 67 00 00 01 00
0060 01 00 00 29 05 ac 00 00 00 00 00 00
Domain Name System: Protocol
Packets: 43 · Displayed: 4 (9.3%) · Dropped: 0 (0.0%)
Profile: Default

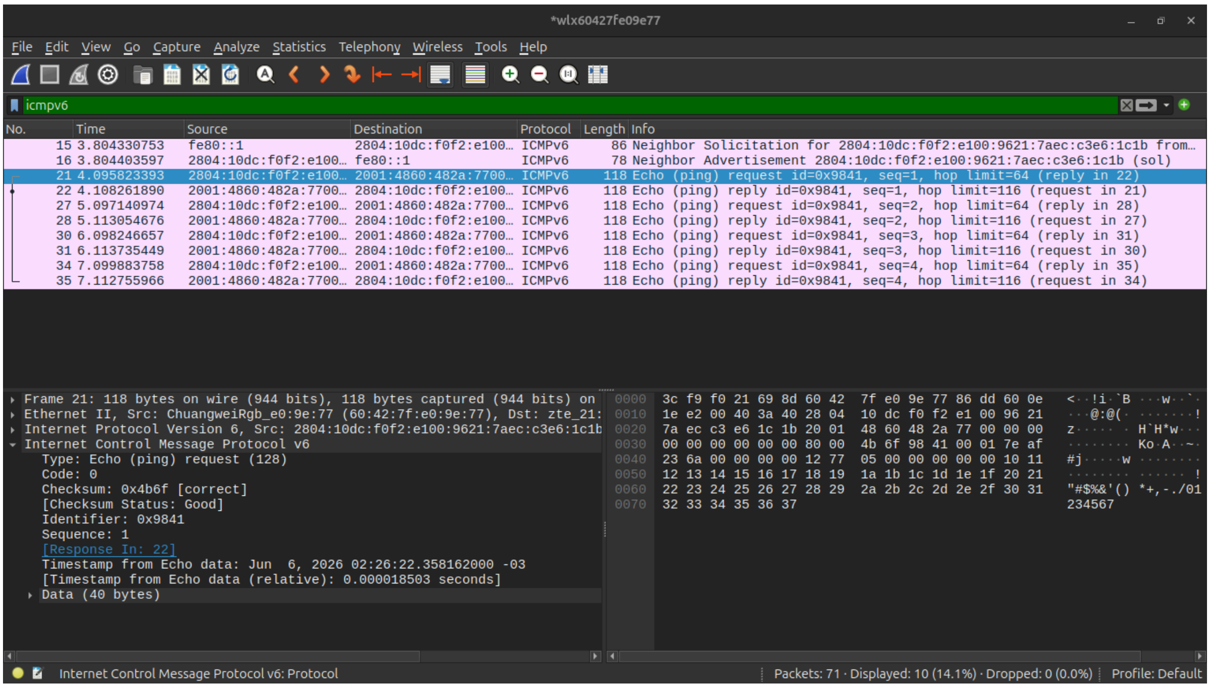

*wlx60427fe09e77
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Length Info
9 2.419494092 2804:10dc:f0f2:e100… 2804:10dc:8100:22::… DNS 108 Standard query 0xd557 A www.wikipedia.org OPT
10 2.427128390 2804:10dc:8100:22::… 2804:10dc:f0f2:e100… DNS 153 Standard query response 0xd557 A www.wikipedia.org CNAME dyna.wikimedi…
11 2.428069983 2804:10dc:f0f2:e100… 2804:10dc:8100:22::… DNS 109 Standard query 0xdde3 AAAA dyna.wikimedia.org OPT
12 2.432695888 2804:10dc:8100:22::… 2804:10dc:f0f2:e100… DNS 137 Standard query response 0xdde3 AAAA dyna.wikimedia.org AAAA 2a02:ec80:…
Questions: 1
Answer RRs: 2
Authority RRs: 0
Additional RRs: 1
Queries
www.wikipedia.org: type A, class IN
Name: www.wikipedia.org
[Name Length: 17]
[Label Count: 3]
Type: A (1) (Host Address)
Class: IN (0x0001)
Answers
www.wikipedia.org: type CNAME, class IN, cname dyna.wikimedia.org
dyna.wikimedia.org: type A, class IN, addr 195.200.68.224
Additional records
[Request In: 9]
[Time: 0.007634298 seconds]
0000 60 42 7f e0 9e 77 3c f9 f0 21 69 8d 86 dd 60 00
0010 00 00 00 63 11 3b 28 04 10 dc 81 00 00 22 00 00
0020 00 00 00 00 00 22 28 04 10 dc f0 f2 e1 00 96 21
0030 7a ec c3 e6 1c 1b 00 35 8c 7d 00 63 9d f8 d5 57
0040 81 80 00 01 00 02 00 00 00 01 03 77 77 77 09 77
0050 69 6b 69 70 65 64 69 61 03 6f 72 67 00 00 01 00
0060 01 c0 0c 00 05 00 01 00 00 88 05 00 11 04 64 79
0070 6e 61 09 77 69 6b 69 6d 65 64 69 61 c0 1a c0 2f
0080 00 01 00 01 00 00 00 80 00 04 c3 c8 44 e0 00 00
0090 29 04 d0 00 00 00 00 00 00
Domain Name System: Protocol
Packets: 43 · Displayed: 4 (9.3%) · Dropped: 0 (0.0%)
Profile: Default

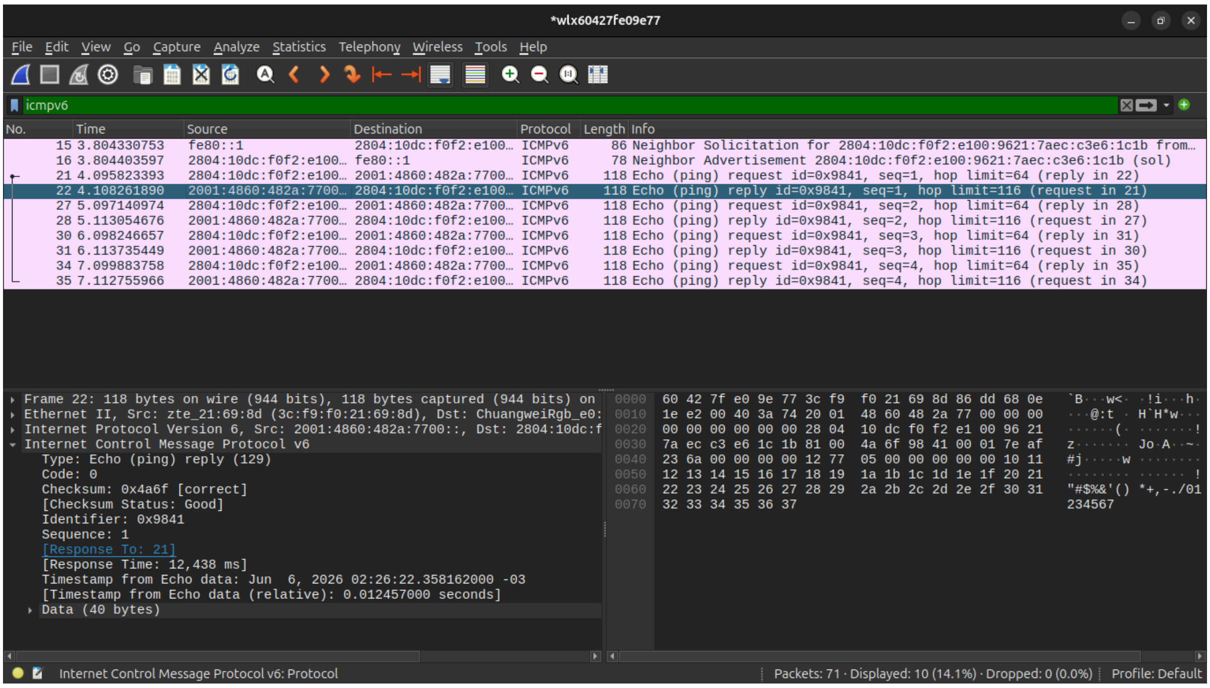


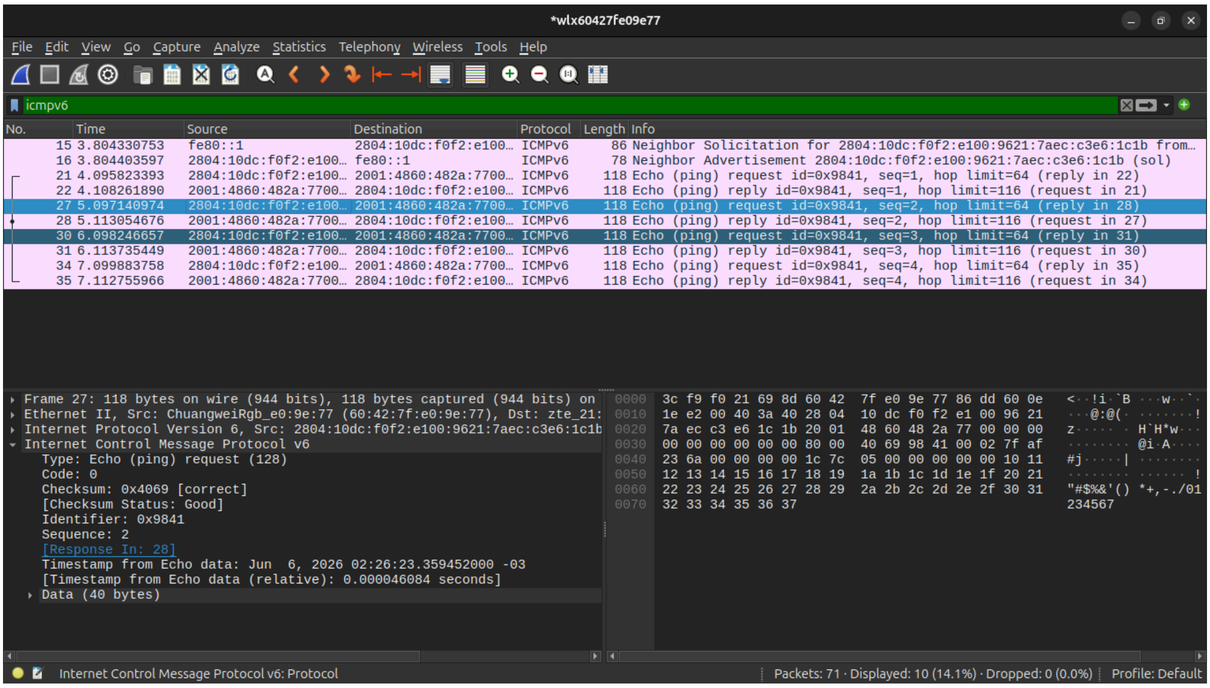


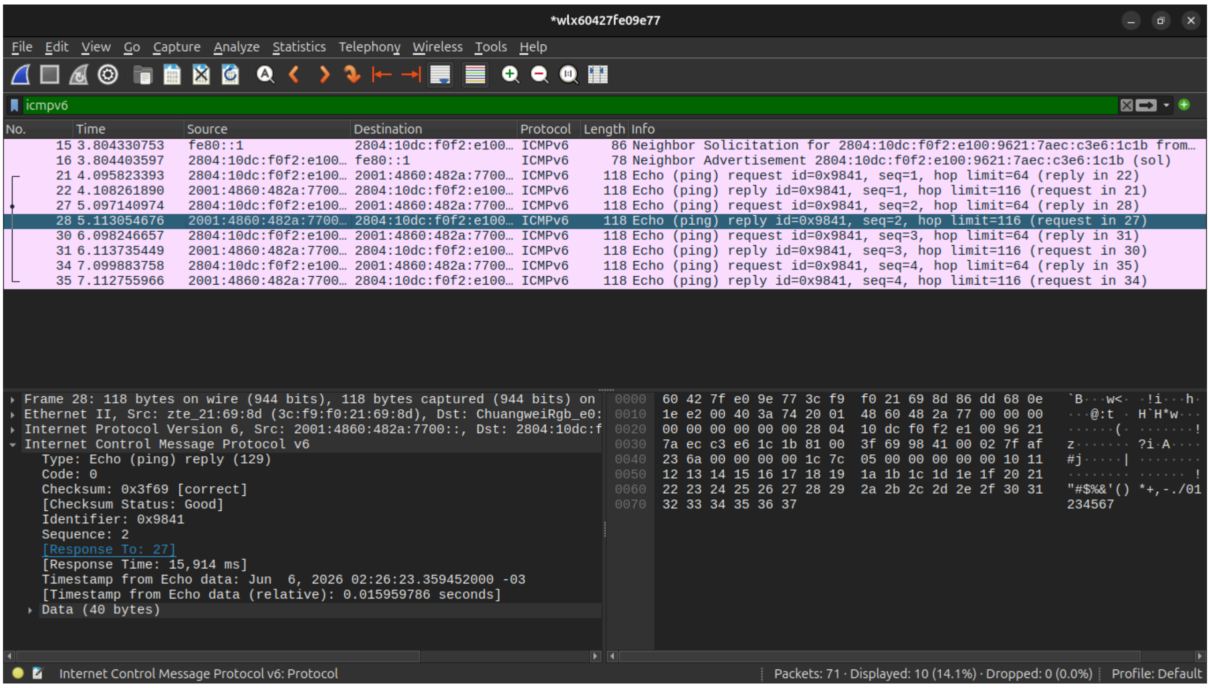
O primeiro pacote mostra a solicitação do Ip do www.wikipedia.org pelo Type A o que mostra ser uma solicitação de IPV4. O retorno do servidor DNS mostra no Answers a resposta que o Ip versão 4 do www.wikipedia.org é 195.200.68.224. Logo em baixo é solicitado o IPV6 do wikipedia pelo Type AAAA, que é enviado pelo servidor DNS logo em seguida.

Exercício 8



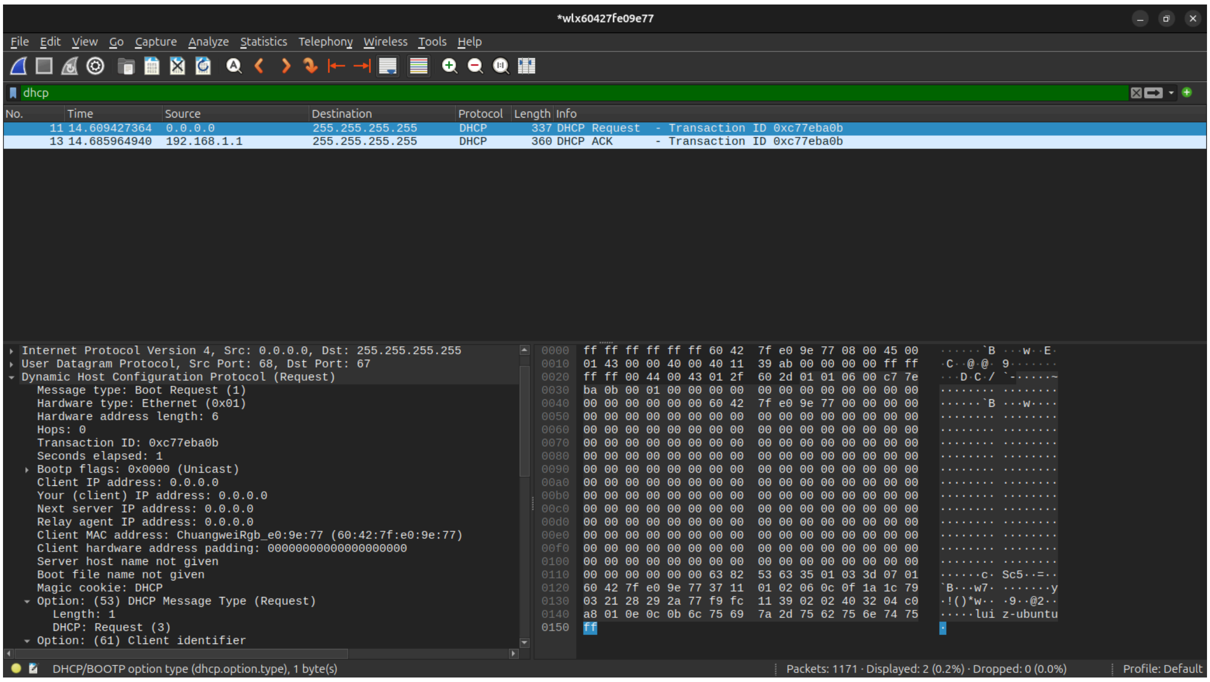


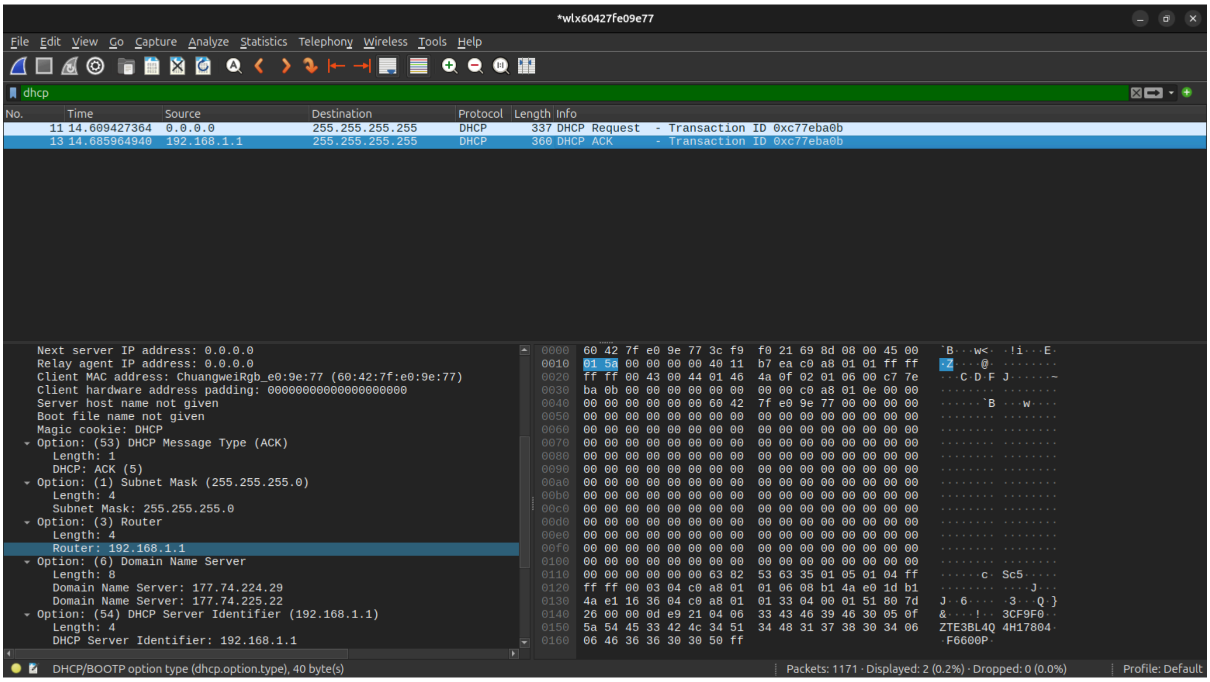




Na linha 21, meu computador envia um pacote Echo request para testar a conexão com o servidor da Google. Olhando os detalhes abaixo, o campo Type está com 128 e o Code com 0, mostrando que é um pedido de teste com o número de sequência 1. Na linha 22, o servidor da Google responde com um pacote Echo reply. Se abrirmos esse pacote, o campo Type muda para 129 (que significa resposta) e o número de sequência continua sendo 1, confirmando que o teste deu certo. Esse processo se repete nas próximas linhas.

Exercício 9





No primeiro pacote ao expandir o Option [53] podemos ver que o meu computador faz uma request e não um discover mostrando que ao desconectar e conectar não houve o processo de descoberta ele apenas enviou uma requisição para reativar o IP antigo. No segundo pacote é devolvido um ack e definido a máscara e router.